

Supplementary Materials for
Comprehensive Profiles of Human Cingulate Gyrus Effective Networks

Phillip Demarest *et al.*

*Corresponding author. Email: leuthardte@wustl.edu

This PDF file includes:

Supplementary Methods
Figs. S1 to S7
Tables S1 to S4

Supplementary Text

Supplemental Methods

Extracting Angular Features of CCEPs. Using the defined onset and offset (CCEP response window) indices (generated from Temporal Dynamics of CC Effective Connectivity Networks in Methods), we derived trial-wise instantaneous phase from the complex broadband signal derived from applying a Hilbert transformation. To increase cross-trial phase consistency and suppress isosyncratic phase excursions, we applied a normalization to align each trial's phase to the across-trial mean at each time point such that $\widetilde{\phi}_i(t) = \cos(\phi_i(t) - \bar{\phi}(t)) \bar{\phi}(t)$.

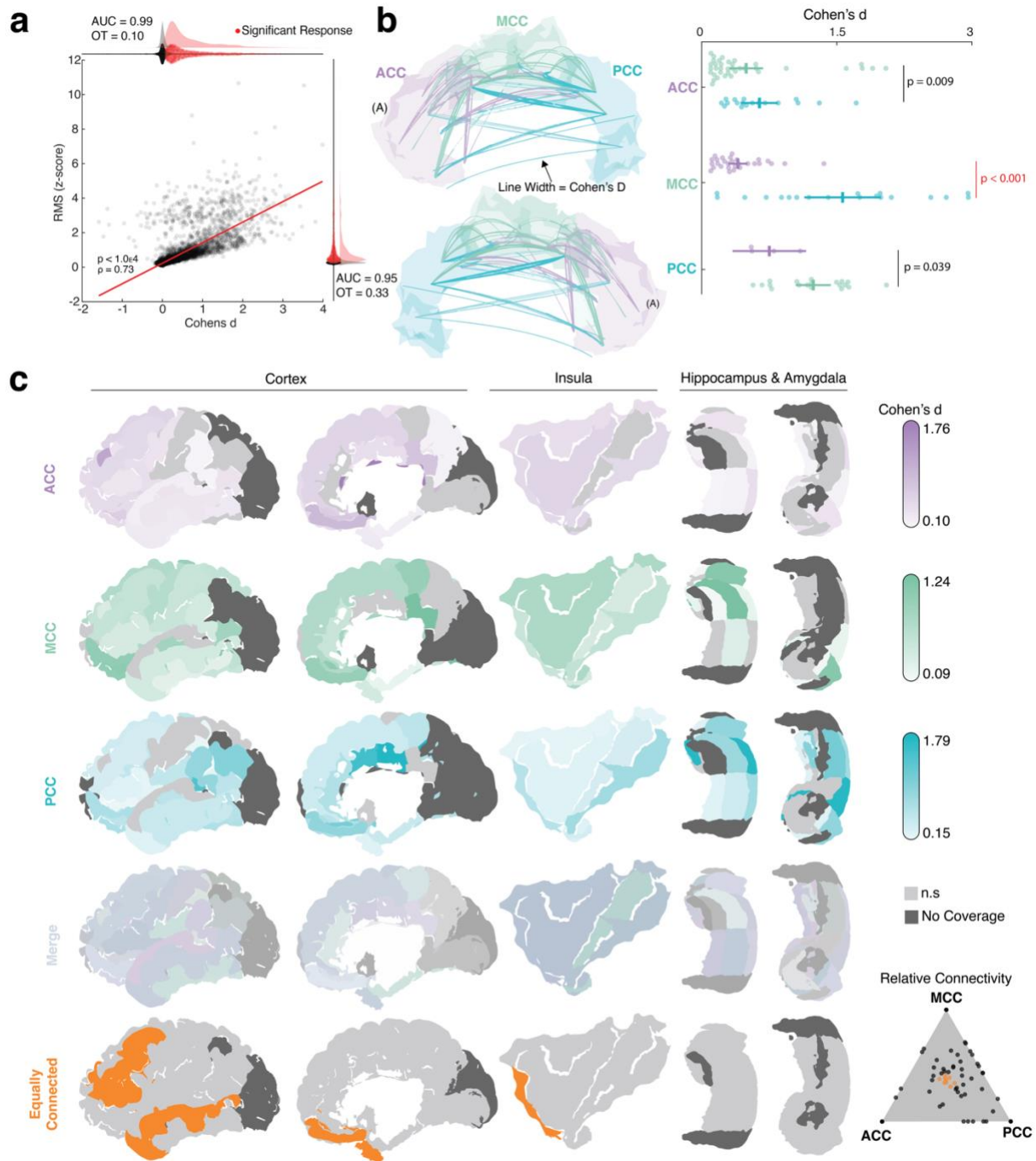
We extracted angular features using a peak-valley characterization constrained to the response window. A single-pass window tracked alternating local maxima and minima of the broadband signal's direction. Each directional changepoint was stored and ranked by absolute amplitude, and the three with the greatest magnitude were maintained and labeled Angle 1, Angle 2, and Angle 3 of the CCEP. Further, we stored the angle of the response onset, the onset of each angle, and the angle of the response offset.

Extracting N1/N2 Features of CCEPs. We extracted CCEP peaks from trial-averaged waveforms using fixed post-SPES windows: 10-50ms for N1 and 60-700ms for N2. A peak was considered present only if its absolute height exceeded a threshold of 5 times the standard deviation of the pre-SPES baseline window. Peaks were found by locating discrete local maxima within the response window. The width of peaks was measured at half-prominence. Prominence was computed from the vertical distance to the lowest contour that connects the peak to higher neighbors.

Supplemental Figure Legends

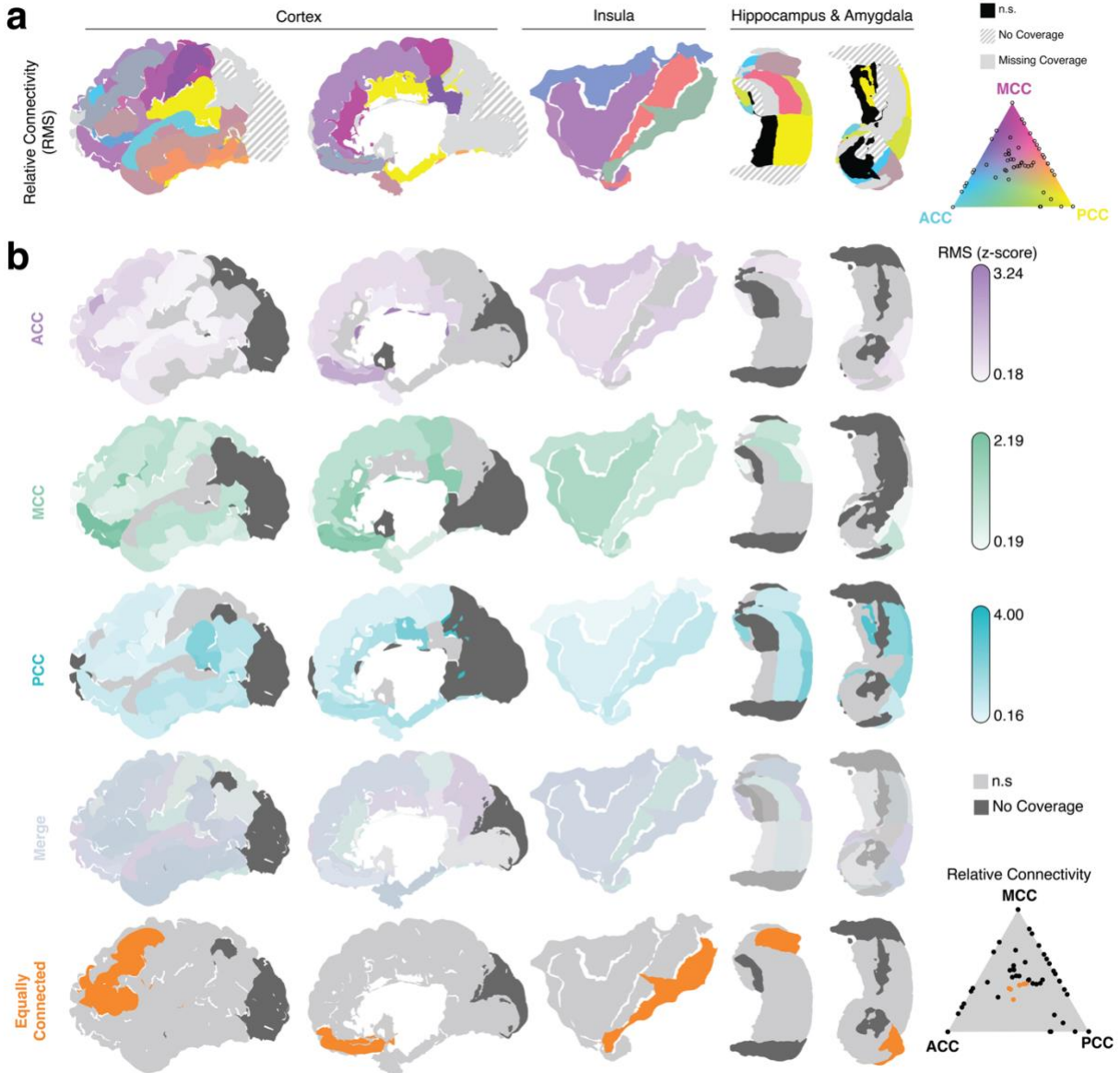


Supplemental Figure 1. Percentage of significant contacts across brain regions. Cortical surface models showing the percentage of significant contacts within each region across all three stimulation conditions (stimulation of the anterior, mid, and posterior cingulate cortices) and both measures of effective connectivity (Cohen's d and RMS).

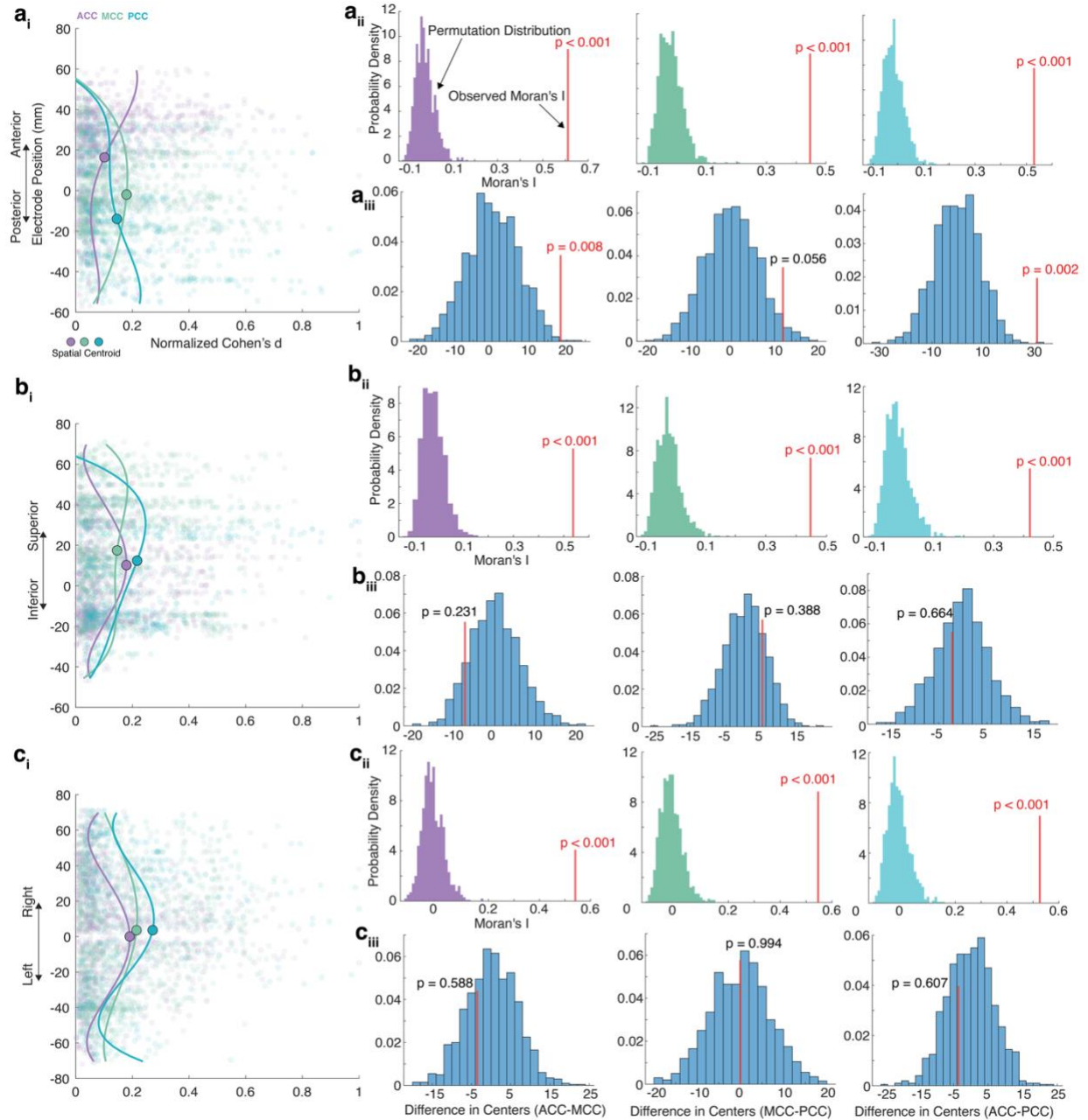


Supplemental Figure 2. Expanded cingulate cortex (CC) effective connectivity modeled using cortico-cortical evoked potential (CCEP) coherence. (a) Relationship between CCEP coherence (Cohen's d) and CCEP amplitude (RMS). The scatterplot shows the relationship between these two measures across all contacts, across all conditions. Distributions of each respective effective connectivity measure projected to the top (Cohen's d) and the right (RMS) axes show significantly connected channels (with respect to each measure) in red. The area under the curve of the receiver operating characteristic was used to determine the performance of classifying significantly effectively connected contacts for each measure. (b) Inter-CC subregion effective

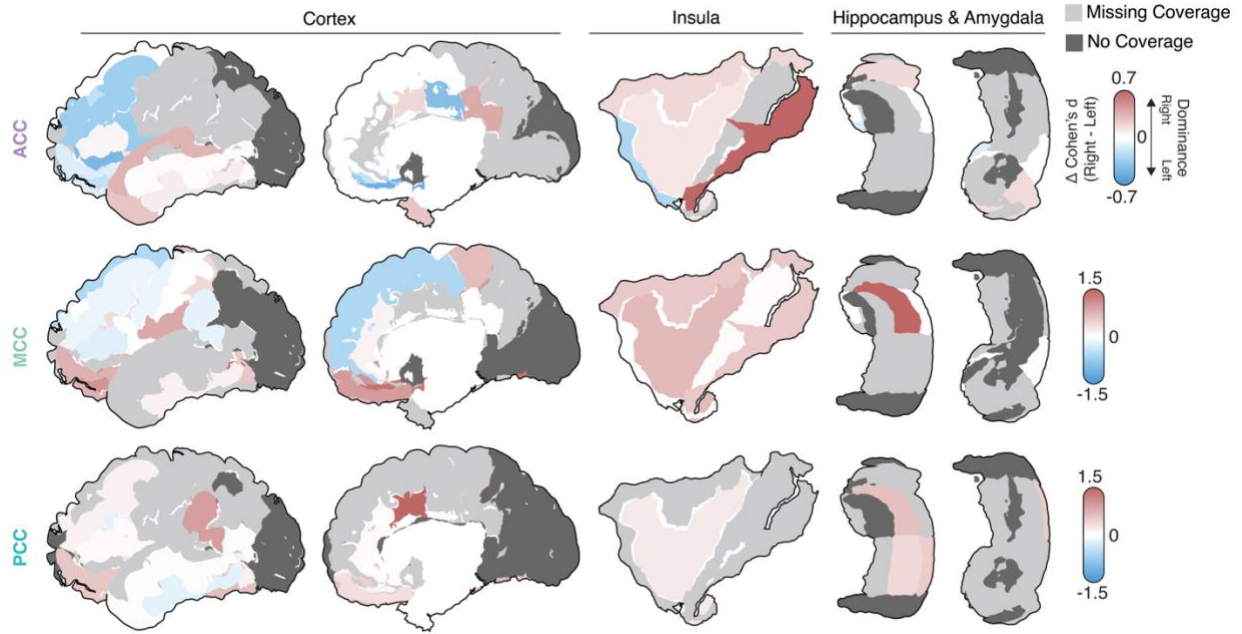
connectivity using CCEP coherence (Cohen's d) obtained from patients implanted with contacts in more than one CC subregion (n = 5 for ACC, n = 6 for MCC, n = 5 for PCC). (Left) Surface model of the CC showing connections between its subregions. Lines were drawn from stimulated contact coordinates to all significantly effectively connected contact coordinates in the other CC subregion(s). Colors of lines represent the stimulated subregion, and line width represents the normalized Cohen's d. (Right) Swarm plot comparing the effective connectivity of each CC subregion with each of the other two subregions. For each stimulation condition, the Cohen's d of contacts located within each of the other non-stimulated subregions was compared using a Wilcoxon rank sum test ($\alpha = 0.01$). Centrality lines show the mean, and the whiskers show the 95% confidence interval. **(c)** (Rows 1-3) Cortical surface models showing effective connectivity, as average Cohen's d, across brain regions for each stimulation condition. (Row 4) Overlay of rows 1-3. (Row 5) Brain regions with highly similar effective connectivity (Cohen's d) to all three CC subregions, determined as any brain regions located within a circle with an area 5% of the triangular plane used to determine relative connectivity to the CC subregions.



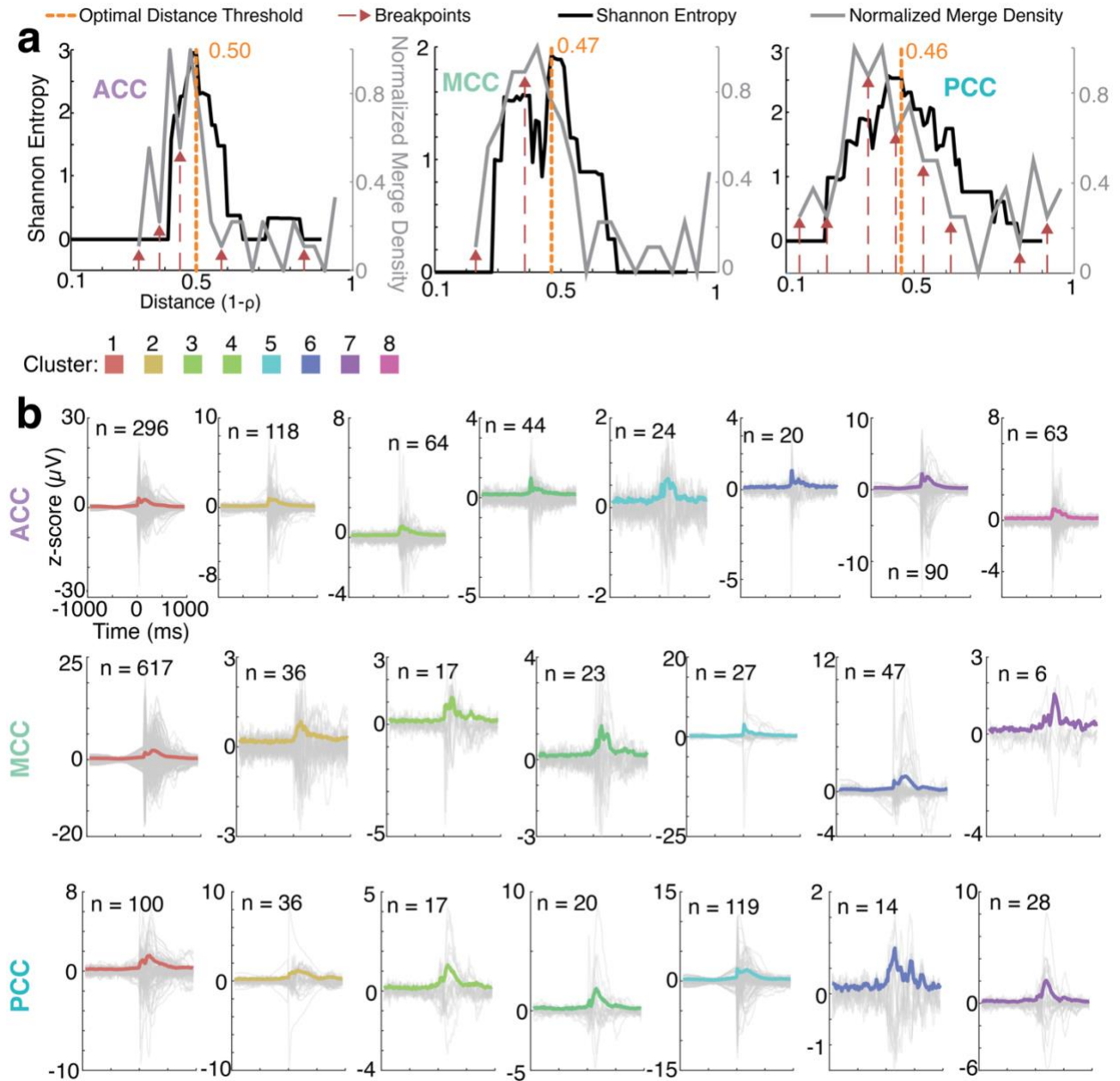
Supplemental Figure 3. Expanded cingulate cortex (CC) effective connectivity modeled using cortico-cortical evoked potential (CCEP) magnitude. (a) Cortical surfaces visualizing the relative effective connectivity, as measured by CCEP RMS, of each CC subregion to each cortical region (See Figure 2 and Methods). (b) (Rows 1-3) Cortical surface models showing effective connectivity, as average CCEP RMS, across brain regions for each stimulation condition. (Row 4) Overlay of rows 1-3. (Row 5) Brain regions with highly similar effective connectivity (RMS) to all three CC subregions, determined as any brain regions located within a circle with an area 5% of the triangular plane used to determine relative connectivity to the CC subregions.



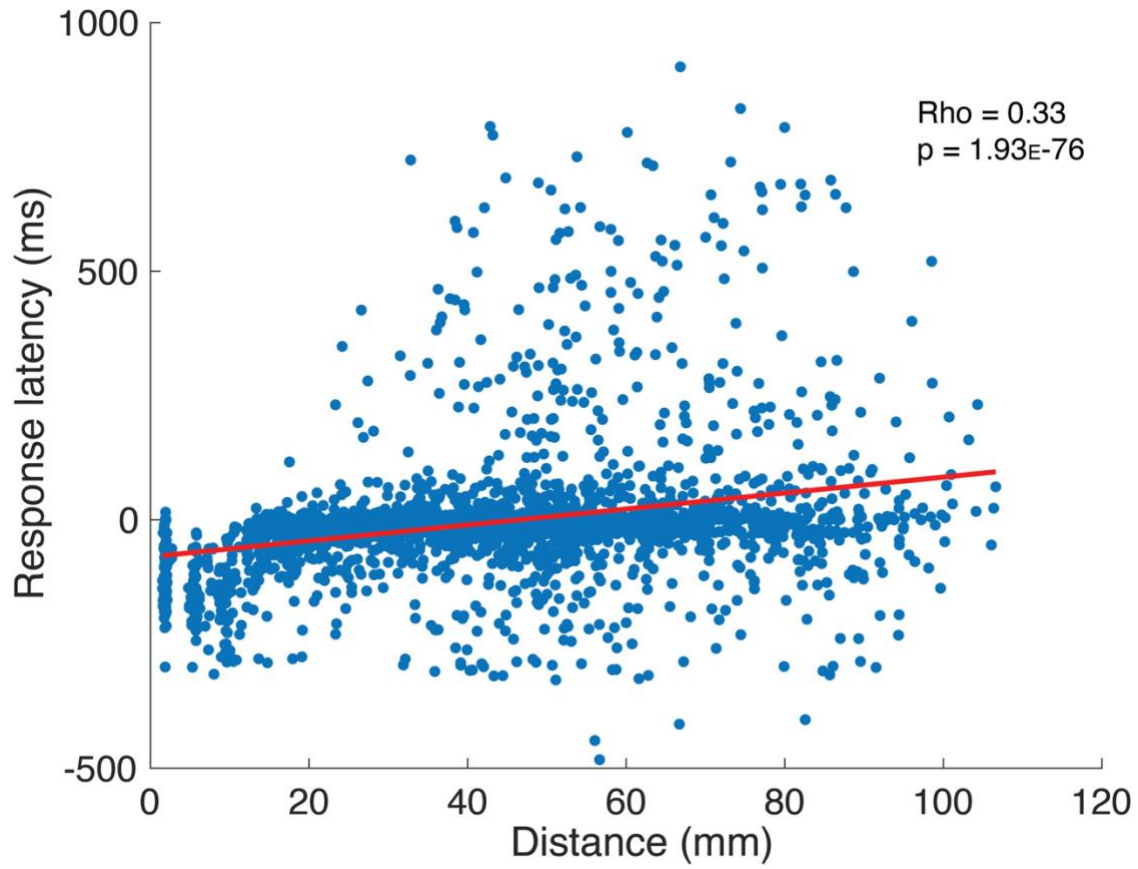
Supplemental Figure 4. Statistics for analyzing the spatial distribution of effective connectivity across the cortex. (a_i) Distributions of normalized effective connectivity (Cohen's d) across the anterior-posterior axis. Lines show a 4th-order polynomial trend line fitted to the mean Cohen's d of 50 evenly spaced bins for each spatial dimension. The circles along the lines represent the centroids of each connectivity distribution. (a_{ii}) Permutation distributions used to determine whether Moran's I for each stimulation condition was significant. (a_{iii}) Permutation-based pairwise comparisons of observed centroid differences were calculated to compare the centroids of distributions for each stimulation condition. (b_i-b_{iii}) Same analysis as in a for the inferior-superior axis. (b_i-b_{iii}) Same analysis as in a for the left-right axis.



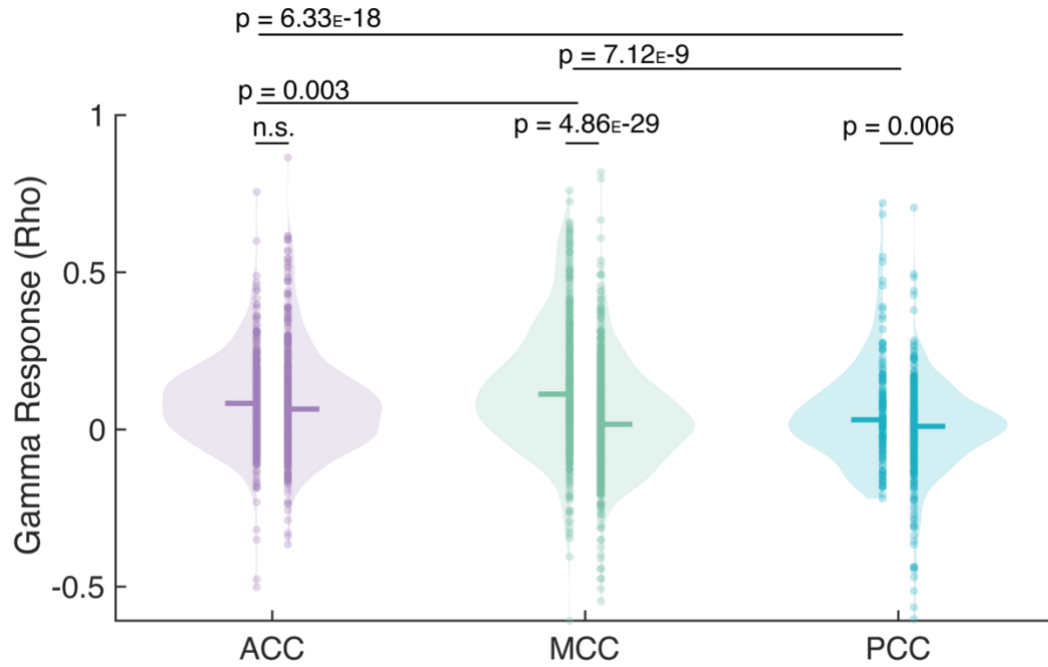
Supplemental Figure 5. Hemispheric differences in effective connectivity across cortical regions. Cortical surface models showing the difference in average Cohen's d between the left and right hemispheres across regions for each stimulation condition. Differences were calculated as average right hemisphere Cohen's d minus average left hemisphere Cohen's d for each brain region. Blue represents regions with greater Cohen's d in the right hemisphere, and red represents greater Cohen's d in the left hemisphere.



Supplemental Figure 6. Hierarchical clustering of cingulate cortex (CC) effective connectivity networks. (a) Optimal distance threshold selection for hierarchical clustering of each CC subregion's effective connectivity network dendrogram. Optimal distance to resolve clusters was calculated using a weighted combination of the Shannon Entropy (left axis) and normalized merge density (right axis) of network dendrograms. Plots show these metrics across all distances of each dendrogram. (b) Visualization of cortico-cortical evoked potential (CCEP) morphology across clusters for each stimulation condition. Colors correspond to each cluster. Grey lines show average CCEPs recorded at each contact within each cluster. Colored lines show the average magnitude of the CCEPs within each cluster (to account for differences in CCEP polarity across contacts).



Supplemental Figure 7. Relationship between CCEP response latency and distance to the stimulation location. Spearman correlation between the detected onset latency (after spectral smearing due to broadband filtering) and the distance of contact to the site of stimulation.



Supplemental Figure 8. Comparisons of Stimulation Evoked Across Conditions and Hemispheres. Gamma responses to stimulation across all contacts between stimulation conditions. Comparisons were also made between hemispheres (left vs right distribution within each condition).

Subj. ID	Age	Length of Stay (Days)	Seizure Onset Zone	Note
BJH024	38	7	multi-focal, deep AR, right temporal pole, unclear onsets, right hippocampal onsets, left hippocampal onset	multifocal epilepticform discharges: bilateral hippocampi, bilateral inferior temporal gyrus, bilateral middle temporal gyrus, bilateral fusiform gyrus, left piriform gyrus, left periform cortex, left perirhinal cortex, left entorhinal cortex and left amygdala
BJH026	40	9	focal seizure ,left ant hippocampus, left posterior hippo, left entorhinal cortex/ anterior parahippocampal gyrus	focal epileptic discharges in left anterior hippocampus (H1-2), left posterior hippocampus (IL1-3), left entorhinal cortex/ anterior parahippocampal gyrus(J1-2), left posterior cingulate cortex (D1-2),left posterior parahippocampal gyrus (K1-5)
BJH030	22	7	focal seizure onsets at right hippocampal body, right hippocampal head, and right amygdala	multi-focal epileptic discharges: bilateral hippocampal region, bilateral amygdala, bilateral parahippocampal gyrus,
BJH031	30	10	right medial temporal lobe	Frequent focal epileptic discharges over bilateral hemisphere, more frequent and prominent in the right hemisphere, and maximal at right posterior hippocampus and right temporal lobe. The background appeared over right subcentral gyrus, dorsal mid insula, anterior thalamus, and right STG, HG, posterior insula, and posterior thalamus.
BJH032	27	9	right hippocampus; left amygdala; left anterior/posterior hippocampus;	Interictal epileptiform discharges in AL1-4, BL1-3, BR1-2, CR4-7; less frequent epileptiform discharges over AR1-2, ER1-4, FR1-4, GR1-8.
BJH035	41	8	left inferior temporal pole, left perirhinal cortex, right anterior temporal pole,	Multifocal epilepticform discharges over left anterior hippocampus, left posterior hippocampus, left amygdala, left perirhinal cortex, left entorhinal cortex, left piriform cortex, right anterior hippocampus, right amygdala, right anterior temporal lobe.
BJH045	22	21	Unidentified	Epileptiform discharges in right supramarginal gyrus, right pre/post central gyrus, right middle frontal gyrus, right superior frontal gyrus, right amygdala, anterior hippocampus.

BJH046	60	14	left posterior MFG, left anterior SFG, left MFG, left posterior SFG	Multi-focal discharges in left MFG/BA46 (HL10-11), left MFG/BA8v (IL 8-9), left anterior SFG/BA9 (ML1-3), and left anterior SFG/BA8, left posterior and anterior SFG/BA9, left hippocampus
BJH050	28	11	right STG, right hippocampus, right parahippocampal gyrus	Frequent epileptiform discharges in right superior temporal gyrus, right anterior/posterior hippocampus and amygdala; Infrequent epileptiform discharges in inferior temporal lobe, entorhinal cortex, and superior temporal lobe.
BJH052	24	8	Left STG, left MTG/STS, left ITG/fusiform gyrus/superior temporal pole, left piriform cortex	Multifocal epileptiform discharges in right anterior hippocampus, left posterior hippocampus, left entorhinal cortex, left STG, left MTG/STS, left ITG/fusiform gyrus, superior temporal pole, left piriform cortex,
BJH053	20	12	right MFG/IFG pars triangularis (lesion), right pars opercularis (lesion), right IFG pars triangularis (lesion)/dorsal anterior insula and right dorsal mid insula.	Focal epileptiform discharges in right MFG/IFG pars triangularis (lesion), right dorsal anterior insula, right IFG pars opercularis/transmantle (lesion)
BJH054	36	4	left MTG, ITG, FuG, STG, HeG, piriform cortex, MFG	Abnormal spike wave discharges in left MTG, ITG, FuG, STG, HeG, piriform cortex, MFG, left anterior and posterior hippocampus, rarely in right amygdala and anterior hippocampus
BJH056	36	10	left parahippocampal gyrus, right collateral sulcus (parahippocampal gyrus/fusiform gyrus junction)	Epileptiform discharges in left/right anterior hippocampus, left/right entorhinal cortex, left/right parahippocampal gyrus, occasionally in left amygdala and posterior hippocampus, and right posterior hippocampus.
BJH058	22	14	Right middle frontal gyrus, right superior frontal gyrus, right anterior hippocampus, right amygdala,	Multi-focal discharges in right middle frontal gyrus, right superior frontal gyrus. Focal epileptiform discharges in right pre-/post-central gyrus, right anterior hippocampus

BJH062	53	6	periventricular cleft and middle/superior frontal gyrus	Frequent epileptiform discharges in left deep cleft, left anterior cleft, occasionally in left superior frontal gyrus, left periventricular cleft, left middle frontal gyrus, left superior frontal gyrus, left middle anterior cleft, left middle anterior cleft, and left superior frontal gyrus .
--------	----	---	---	--

Supplemental Table 1. Patient Demographics.

Region	Condition	Ipsi Difference	Direction Change Ipsi	Ipsi Significance (p-Value)	Ipsi Right Count	Ipsi Left Count
Amygdala	ACC	-0.006327476	No	0.963585434	12	6
Frontal Lobe	ACC	-0.05520696	No	0.005463846	101	76
Hippocampus	ACC	-0.130179078	No	0.021328671	9	7
Insula	ACC	-0.006286412	No	0.759547251	50	31
Operculum	ACC	-0.05167328	Yes	0.133333333	4	2
Orbitofrontal cortex	ACC	0.016996539	No	0.768661087	76	68
Parietal Lobe	ACC				0	4
Somato-Motor Cortex	ACC				0	8
Temporal Lobe	ACC	-0.034851422	No	0.200055286	54	42
Thalamus	ACC				0	0
Amygdala	MCC				13	0
Frontal Lobe	MCC	-0.140119716	No	8.38433E-07	114	189
Hippocampus	MCC	-0.091411734	Yes	0.063636364	9	3
Insula	MCC	-0.122916087	No	5.61487E-05	53	50
Operculum	MCC	-0.058628159	No	0.077064725	43	25

Orbitofrontal cortex	MCC	-0.154607115	Yes	5.5968E-08	42	30
Parietal Lobe	MCC	-0.10378725	No	0.042018858	6	18
Somato-Motor Cortex	MCC	0.030710147	No	0.289312945	81	85
Temporal Lobe	MCC	-0.161233568	No	0.004149393	70	12
Thalamus	MCC				16	0
Amygdala	PCC				12	0
Frontal Lobe	PCC	0.218860357	Yes	3.38945E-05	12	38
Hippocampus	PCC	-0.000962639	Yes	0.066353899	17	4
Insula	PCC	-0.108074975	Yes	0.05994006	9	5
Operculum	PCC	-0.095082488	Yes	0.088888889	8	2
Orbitofrontal cortex	PCC	0.011897799	Yes	0.089348446	23	16
Parietal Lobe	PCC	0.054785631	Yes	0.683516484	4	12
Somato-Motor Cortex	PCC				0	4
Temporal Lobe	PCC	-0.070645464	Yes	0.014640058	136	40
Thalamus	PCC				8	0

Supplemental Table 2. Hemispheric Differences in Gamma by Region. Ipsi-Difference: The magnitude difference in gamma response between left and right hemispheres (right vs left). Directional Change Ipsi: Whether there was a difference in the direction of the evoked gamma response (i.e. increase vs decrease). Ipsi Right/Left Count: number of channels included in comparison.

Region	Condition	Avg Gamma Response, Right Hemisphere (Rho)	Avg Gamma Response, Left Hemisphere (Rho)
Amygdala	ACC	0.06908728	0.075414756
Frontal Lobe	ACC	0.03848938	0.09369634
Hippocampus	ACC	0.060897112	0.191076191
Insula	ACC	0.028705081	0.034991492
Operculum	ACC	-0.002980879	0.054654159
Orbitofrontal cortex	ACC	0.142546805	0.125550266
Parietal Lobe	ACC		0.104112489
Somato-Motor Cortex	ACC		-0.026907005
Temporal Lobe	ACC	0.088333385	0.123184807
Thalamus	ACC		
Amygdala	MCC	-4.73255E-05	
Frontal Lobe	MCC	0.044393965	0.184513682
Hippocampus	MCC	-0.015172522	0.106584256
Insula	MCC	0.044163829	0.167079916
Operculum	MCC	0.048140403	0.106768562
Orbitofrontal cortex	MCC	-0.070059829	0.224666945
Parietal Lobe	MCC	0.008044952	0.111832201
Somato-Motor Cortex	MCC	0.064485673	0.033775526
Temporal Lobe	MCC	0.021047709	0.182281277
Thalamus	MCC	0.004035729	
Amygdala	PCC	0.059392992	
Frontal Lobe	PCC	-0.276615081	0.057754724
Hippocampus	PCC	0.055174608	-0.056137247
Insula	PCC	0.003382583	-0.111457558
Operculum	PCC	-0.069548657	0.164631145
Orbitofrontal cortex	PCC	0.037062377	-0.025164578
Parietal Lobe	PCC	-0.055368065	0.000582435
Somato-Motor Cortex	PCC		-0.002570347
Temporal Lobe	PCC	-0.000363677	0.071009141
Thalamus	PCC	-0.026323722	

Supplemental Table 3. Average Gamma Responses by Region and Hemisphere.

Name	Region	Class
Accessory-Basal-nucleus	<i>Accessory Basal Nucleus of the Amygdala</i>	Amygdala
Anterior-amygdaloid-area-AAA	<i>Anterior Amygdaloid Area</i>	Amygdala
Basal-nucleus	<i>Basal Nucleus of the Amygdala</i>	Amygdala
Corticoamygdaloid-transitio	<i>Corticoamygdaloid transition area</i>	Amygdala
Lateral-nucleus	<i>Lateral nucleus of the amygdala</i>	Amygdala
Paralaminar-nucleus	<i>Paralaminar nucleus of the amygdala</i>	Amygdala
G_and_S_cingul-Ant	Anterior part of the <i>cingulate gyrus and sulcus</i> (ACC)	Cingulate cortex
G_and_S_cingul-Mid-Ant	Middle-anterior part of the <i>cingulate gyrus and sulcus</i> (aMCC)	Cingulate cortex
G_and_S_cingul-Mid-Post	Middle-posterior part of the <i>cingulate gyrus and sulcus</i> (pMCC)	Cingulate cortex
G_cingul-Post-dorsal	Posterior-dorsal part of the <i>cingulate gyrus</i> (dPCC)	Cingulate cortex
G_cingul-Post-ventral	Posterior-ventral part of the cingulate gyrus (vPCC, <i>isthmus of the cingulate gyrus</i>)	Cingulate cortex
S_cingul-Marginalis	<i>Marginal branch</i> (or part) <i>of the cingulate sulcus</i>	Cingulate cortex
G_and_S_frontomargin	Fronto-marginal gyrus (of Wernicke) and sulcus	Frontal Lobe
G_and_S_transv_fronto pol	Transverse frontopolar gyri and sulci	Frontal Lobe
G front inf-Opercular	<i>Opercular part of the inferior frontal gyrus</i>	Frontal Lobe
G front inf-Orbital	<i>Orbital part of the inferior frontal gyrus</i>	Frontal Lobe
G front inf-Triangul	<i>Triangular part of the inferior frontal gyrus</i>	Frontal Lobe
G front middle	<i>Middle frontal gyrus</i> (F2)	Frontal Lobe
G front sup	<i>Superior frontal gyrus</i> (F1)	Frontal Lobe
S front inf	<i>Inferior frontal sulcus</i>	Frontal Lobe
S front middle	Middle frontal sulcus	Frontal Lobe
S front sup	<i>Superior frontal sulcus</i>	Frontal Lobe
CA1-body	<i>Body of the Cornu Ammonis Area 1</i>	Hippocampus
CA1-head	<i>Head of the Cornu Ammonis Area 1</i>	Hippocampus
CA3-head	<i>Head of the Cornu Ammonis Area 3</i>	Hippocampus
CA4-body	<i>Body of the polymorphic layer of the dentate gyrus</i>	Hippocampus
CA4-head	<i>Head of the polymorphic layer of the dentate gyrus</i>	Hippocampus
GC-ML-DG-body	<i>Body of the granule cell and molecular layer of the dentate gyrus</i>	Hippocampus

GC-ML-DG-head	<i>Head of the granule cell and molecular layer of the dentate gyrus</i>	Hippocampus
HATA	<i>Hippocampal-amygdaloid transition area</i>	Hippocampus
hippocampal fissure	<i>Hippocampal Fissure</i>	Hippocampus
molecular_layer_HP-body	<i>Body of the molecular layer of the hippocampus</i>	Hippocampus
molecular_layer_HP-head	<i>Head of the molecular layer of the hippocampus</i>	Hippocampus
presubiculum-body	<i>Body of the presubiculum</i>	Hippocampus
subiculum-body	<i>Body of the subiculum</i>	Hippocampus
subiculum-head	<i>Head of the subiculum</i>	Hippocampus
G_Ins_lg_and_S_cent_ins	<i>Long insular gyrus and central sulcus of the insula</i>	Insula
G_insular_short	<i>Short insular gyri</i>	Insula
S_circular_insula_ant	Anterior segment of the <i>circular sulcus of the insula</i>	Insula
S_circular_insula_inf	Inferior segment of the <i>circular sulcus of the insula</i>	Insula
S_circular_insula_sup	Superior segment of the <i>circular sulcus of the insula</i>	Insula
G_precentral	<i>Precentral gyrus</i>	Motor Cortex
S_precentral-inf-part	Inferior part of the <i>precentral sulcus</i>	Motor Cortex
S_precentral-sup-part	Superior part of the <i>precentral sulcus</i>	Motor Cortex
G_and_S_occipital_inf	Inferior occipital gyrus (O3) and sulcus	Occipital Lobe
G_cuneus	<i>Cuneus</i> (O6)	Occipital Lobe
G_occipital_middle	Middle occipital gyrus (O2, lateral occipital gyrus)	Occipital Lobe
G_occipital_sup	Superior occipital gyrus (O1)	Occipital Lobe
Pole_occipital	<i>Occipital pole</i>	Occipital Lobe
S_calcarine	<i>Calcarine sulcus</i>	Occipital Lobe
S_collat_transv_post	Posterior transverse collateral sulcus	Occipital Lobe
S_oc_middle_and_Lunatus	Middle occipital sulcus and lunatus sulcus	Occipital Lobe
S_oc_sup_and_transversal	Superior occipital sulcus and <i>transverse occipital sulcus</i>	Occipital Lobe
S_occipital_ant	Anterior occipital sulcus and <i>preoccipital notch</i> (temporo-occipital incisure)	Occipital Lobe

S parieto occipital	<i>Parieto-occipital sulcus</i> (or fissure)	Occipital Lobe
G and S subcentral	Subcentral gyrus (central operculum) and sulci	Operculum
Lat Fis-ant-Horizont	Horizontal ramus of the <i>anterior segment of the lateral sulcus</i> (or fissure)	Operculum
Lat Fis-ant-Vertical	Vertical ramus of the <i>anterior segment of the lateral sulcus</i> (or fissure)	Operculum
Lat Fis-post	<i>Posterior ramus</i> (or segment) <i>of the lateral sulcus</i> (or fissure)	Operculum
G orbital	<i>Orbital gyri</i>	Orbitofrontal cortex
G rectus	<i>Straight gyrus</i> , Gyrus rectus	Orbitofrontal cortex
S orbital lateral	Lateral orbital sulcus	Orbitofrontal cortex
S orbital med-olfact	Medial orbital sulcus (<i>olfactory sulcus</i>)	Orbitofrontal cortex
S orbital-H Shaped	<i>Orbital sulci</i> (H-shaped sulci)	Orbitofrontal cortex
S suborbital	Suborbital sulcus (sulcus rostrales, supraorbital sulcus)	Orbitofrontal cortex
G subcallosal	<i>Subcallosal area, subcallosal gyrus</i>	Other
S pericallosal	Pericallosal sulcus (<i>S of corpus callosum</i>)	Other
Cerebellum-Cortex		Other
Inf-Lat-Vent		Other
Lateral-Ventricle		Other
Pallidum		Other
Putamen		Other
VentralDC		Other
Unknown		Other
unknown		Other
G and S paracentral	<i>Paracentral lobule and sulcus</i>	Parietal Lobe
G pariet inf-Angular	<i>Angular gyrus</i>	Parietal Lobe
G pariet inf-Supramar	<i>Supramarginal gyrus</i>	Parietal Lobe
G parietal sup	<i>Superior parietal lobule</i> (lateral part of P1)	Parietal Lobe
G precuneus	<i>Precuneus</i> (medial part of P1)	Parietal Lobe
S interm prim-Jensen	Sulcus intermedius primus (of Jensen)	Parietal Lobe
S_intrapariet_and_P_trans	<i>Intraparietal sulcus</i> (interparietal sulcus) and transverse parietal sulci	Parietal Lobe
S subparietal	<i>Subparietal sulcus</i>	Parietal Lobe
G_postcentral	<i>Postcentral gyrus</i>	Somatosensory Cortex

S_central	<i>Central sulcus</i> (Rolando's fissure)	Somatosensory Cortex
S_postcentral	<i>Postcentral sulcus</i>	Somatosensory Cortex
G_oc-temp_lat-fusifor	<i>Lateral occipito-temporal gyrus</i> (fusiform gyrus, O4-T4)	Temporal Lobe
G_oc-temp_med-Lingual	<i>Lingual gyrus</i> , lingual part of the <i>medial occipito-temporal gyrus</i> , (O5)	Temporal Lobe
G_oc-temp_med-Parahip	<i>Parahippocampal gyrus</i> , parahippocampal part of the <i>medial occipito-temporal gyrus</i> , (T5)	Temporal Lobe
G_temp_sup-G_T_transv	<i>Anterior transverse temporal gyrus</i> (of Heschl)	Temporal Lobe
G_temp_sup-Lateral	Lateral aspect of the <i>superior temporal gyrus</i>	Temporal Lobe
G_temp_sup-Plan_polar	Planum polare of the <i>superior temporal gyrus</i>	Temporal Lobe
G_temp_sup-Plan_tempo	<i>Planum temporale or temporal plane</i> of the superior temporal gyrus	Temporal Lobe
G_temporal_inf	<i>Inferior temporal gyrus</i> (T3)	Temporal Lobe
G_temporal_middle	<i>Middle temporal gyrus</i> (T2)	Temporal Lobe
Pole_temporal	<i>Temporal pole</i>	Temporal Lobe
S_collat_transv_ant	Anterior transverse collateral sulcus	Temporal Lobe
S_oc-temp_lat	Lateral occipito-temporal sulcus	Temporal Lobe
S_oc-temp_med_and_Lingual	Medial occipito-temporal sulcus (<i>collateral sulcus</i>) and lingual sulcus	Temporal Lobe
S_temporal_inf	<i>Inferior temporal sulcus</i>	Temporal Lobe
S_temporal_sup	<i>Superior temporal sulcus</i> (parallel sulcus)	Temporal Lobe
S_temporal_transverse	<i>Transverse temporal sulcus</i>	Temporal Lobe
Pole_temporal	The Temporal pole	Temporal Lobe
Thalamus	Thalamus	Thalamus
Cerebral-White-Matter		White matter
WM-hypointensities		White matter

Supplemental Table 4. Atlas Legend. Region Labels, Region Name, and Region Class definitions used in the present study.